CISS240: Introduction to Programming
Quiz q2002

Name: eagraham1@cougars.ccis.edu Score:

This is a closed-book, no compiler, 5 minute quiz.

Q1. Complete the following function that accepts an array and a length, and overwrites the negative values in the array with 0. For instance if the function is called with the array `{-1, 2, -3, -4, 5, 6, -7, 4}` and 8, on returning from the function, the array becomes `{0, 2, 0, 0, 5, 6, 0, 4}`.
Answer:

```
void clear_negative(double x[], int len)
{
    for (int i = 0; i < len; ++i)
    {
        if (x[i] < 0)
        {
            x[i] = 0;
        }
    }
}
```

Q2. Complete the following function that accepts a 2D array `x` and `n` (at most 1024) and fill `x` with values so that for instance if `n` is `5`, `x` is

```
1 1 1 1 1
1 0 0 0 1
1 0 0 0 1
1 0 0 0 1
1 1 1 1 1
```

i.e., `1`s on the boundary of the 5×5 area surrounding `0`s. (Top left corresponding to row 0, column 0 of the array.)
Answer:

```
void f(int x[1024][1024], int n)
{
    for (int i = 0; i < n; ++i)
    {
        for (int j = 0; j < n; ++j)
        {
            if (i == 0 || i == n - 1 || j == 0 || j == n - 1)
            {
```

```
                x[i][j] = 1;
            }
            else
            {
                x[i][j] = 0;
            }
        }
    }
}
```

Q3. What is the output (or write ERROR).

```
int x[10][10] = {0};

for (int i = 0; i < 10; i++)
{
    x[i][0] = 1;
    x[0][i] = 1;
}

for (int i = 1; i < 10; i++)
{
    for (int j = 1; j < 10; j++)
    {
        x[i][j] = x[i - 1][j] + x[i][j - 1] + x[i - 1][j - 1];
    }
}

std::cout << x[1][2] << '\n';
```

Answer:

```
5
```

Q4. What is the output (or write ERROR).

```
#include <iostream>

void set_to_zero(int x)
{
    x = 0;
}

void set_to_one(int & x)
{
    x = 1;
}

int main()
{
    int x = 42;
    set_to_zero(x);
    std::cout << x << ' ';
    set_to_one(x);
    std::cout << x << '\n';
    return 0;
}
```

Answer:

```
42 1
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.